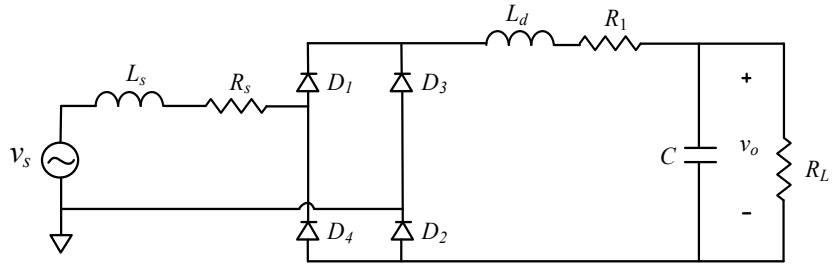
Power Electronics

Lab 2 – Rectifier (con't)

Task1: **Single phase diode bridge rectifier**

Objective: This experiment investigates the effects of inductor on the ac side and large capacitor on the dc side

Build the single-phase diode rectifier as follows



The parameters of the circuits are given as:

$v_s(t)$ = 100sin(2*pi*50) (V)

L_s = 0.1mH, 0.5mH, 1mH

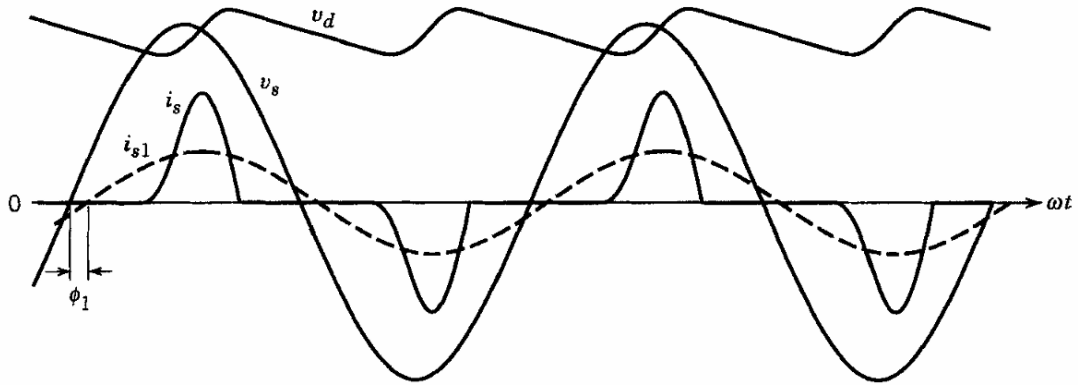
R_s = 0.25Ω

L_d = 1μH

R_1 = 10m

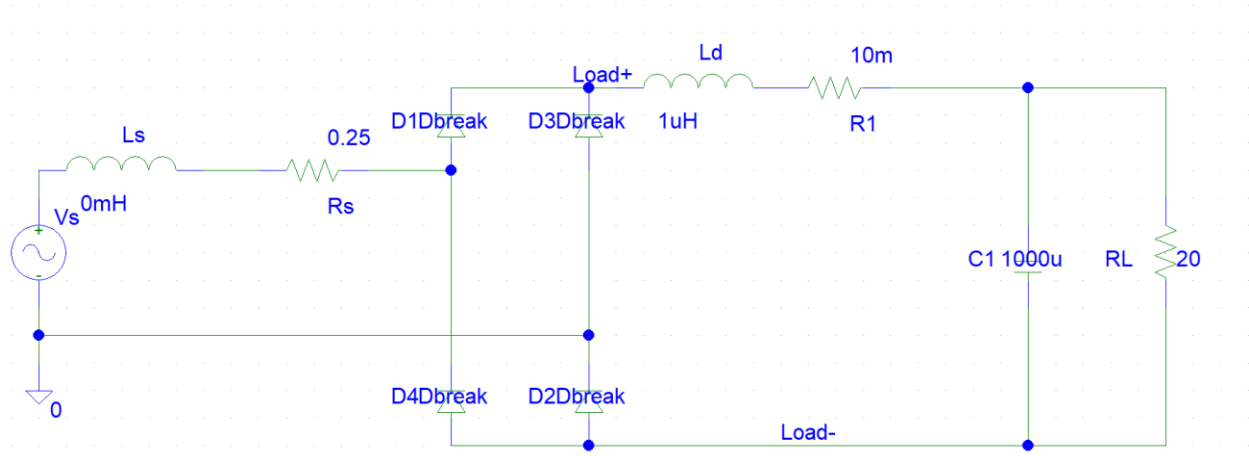
C = 220μF, 470μF,1000μF

R_L = 20Ω

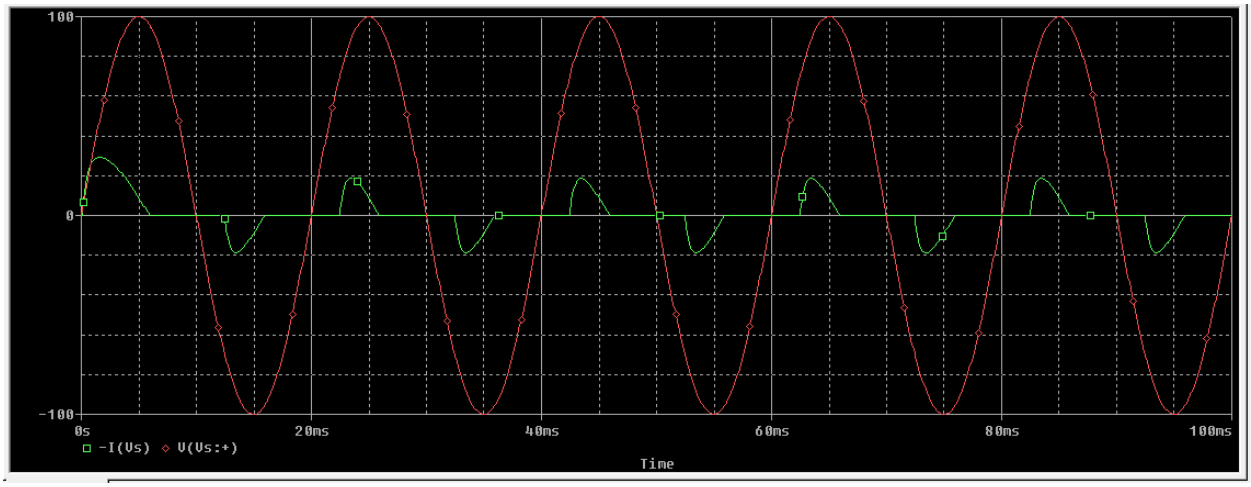


a) Keep the dc side capacitance value as 1000μF, plot the input current waveforms for two cases: L_s = 0 and L_s = 1mH and comment on the effect of L_s

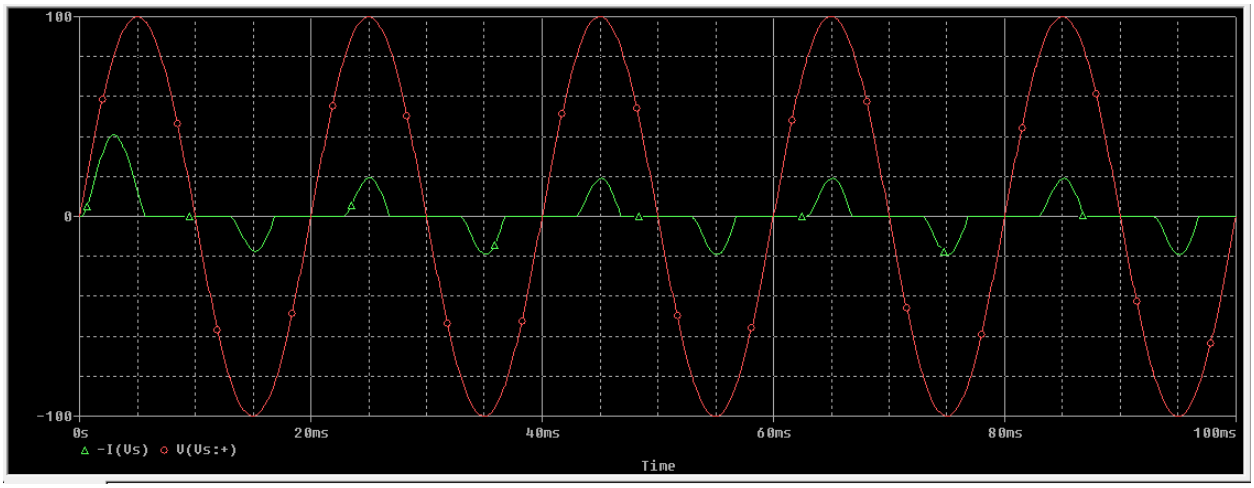
(insert plots here)


Ld
10m
Load+
Ls
0.25
D1Dbreak
D3Dbreak
1uH
R1
0mH
Vs
Rs
C1 1000u
RL
20
0
D4Dbreak
D2Dbreak
Load-

L_s = 0mH


100
0
-100
0s
20ms
40ms
60ms
80ms
100ms
-I(Vs) V(Vs:+)
Time

L_s = 1mH


100
0
-100
0s
20ms
40ms
60ms
80ms
100ms
-I(Vs) V(Vs:+)
Time

Comment:

At Ls = 0 mH: The input current consists of very narrow, high-amplitude pulses. At Ls = 1mH: The introduction of source inductance smoothes the current waveform.

---

b) Keep the dc side capacitance value as 1000μF and change the ac side inductance values to be 0.1mH, 0.5mH, and 1mH.

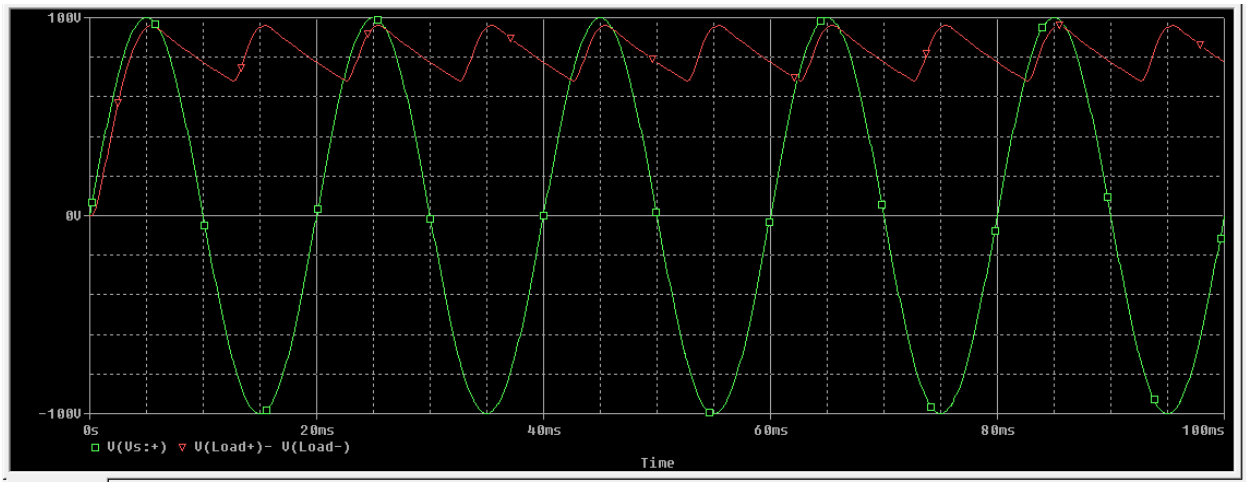
- Plot the output voltages for the three values of the ac side inductance

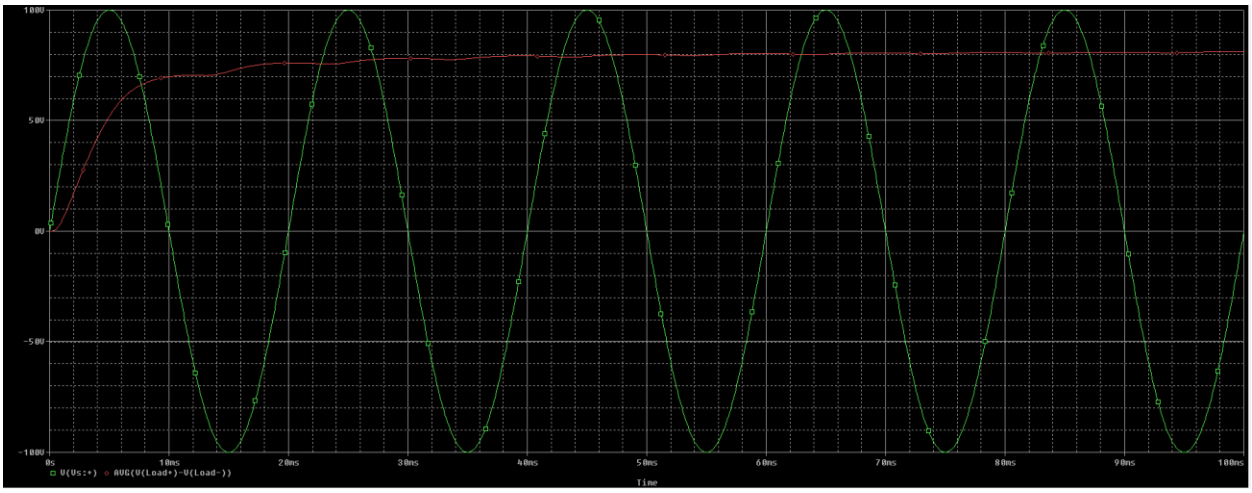
(insert plots here)

- Measure the average values of the output voltages for the three values of ac side inductance

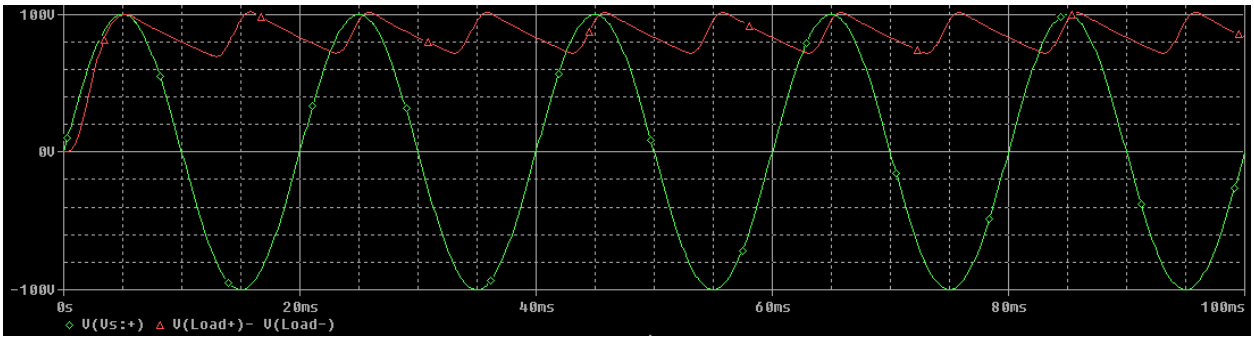
  (To measure the average values, select “Simulate”, then “Add traces”, and then “AVG()”, type the output variable at the “Trace Expression”)

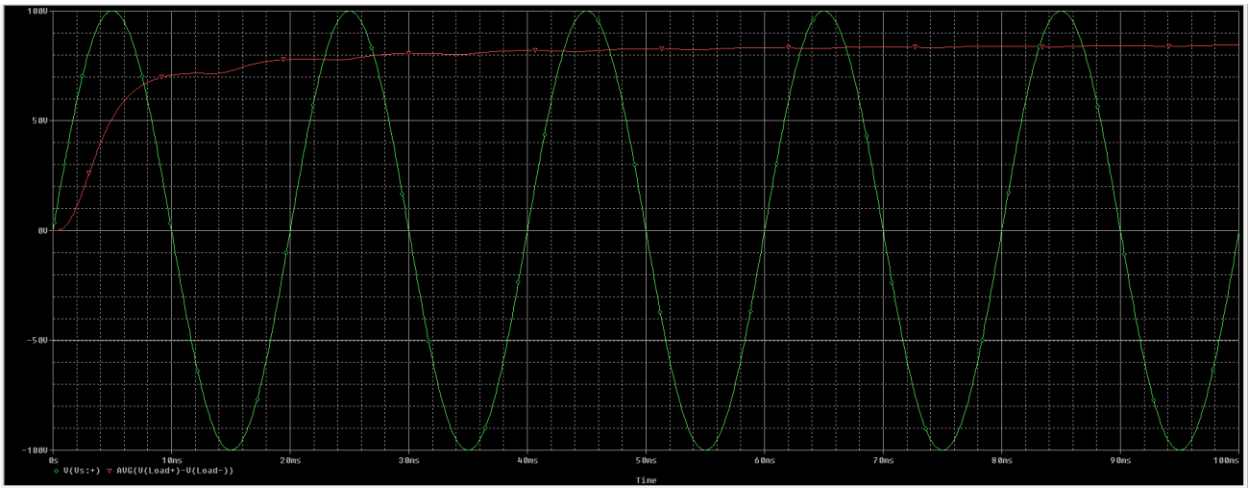
L_s = 0.1mH



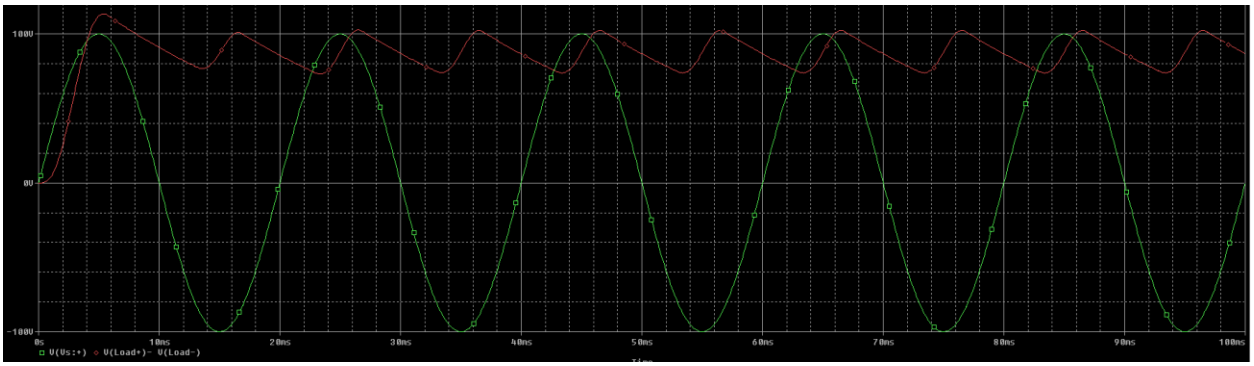


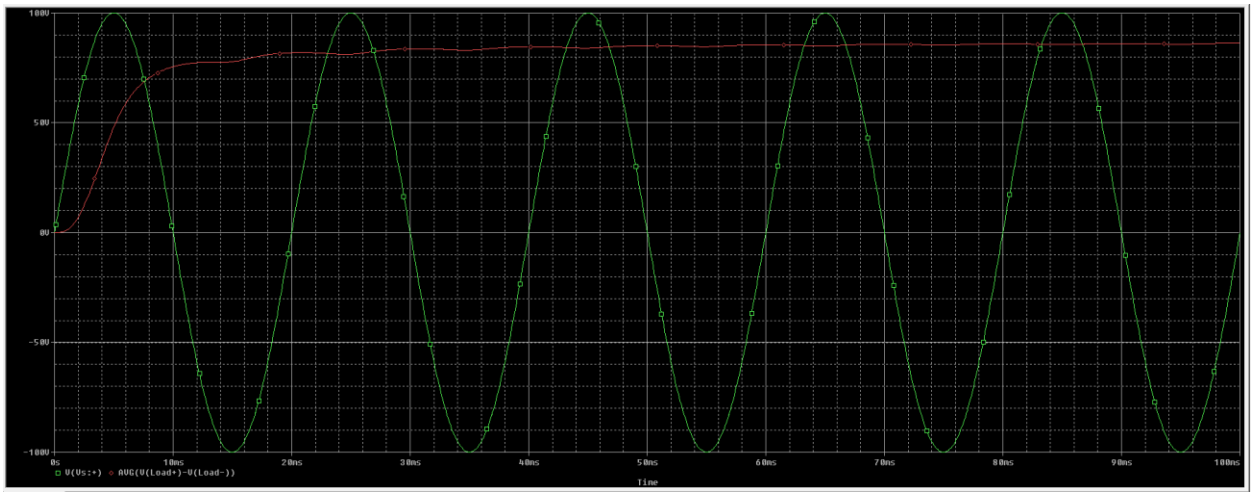
L_s = 0.5mH





L_s = 1mH





Comment:

As the source inductance Ls increases, the average DC output voltage decreases.

The plots show the average voltage line settling at slightly lower levels as Ls increases from 0.1mH to 1mH.

c) Keep the ac side inductance value as 1mH and change the dc side capacitance values to be 220µF, 470µF,1000µF. Repeat part 2.

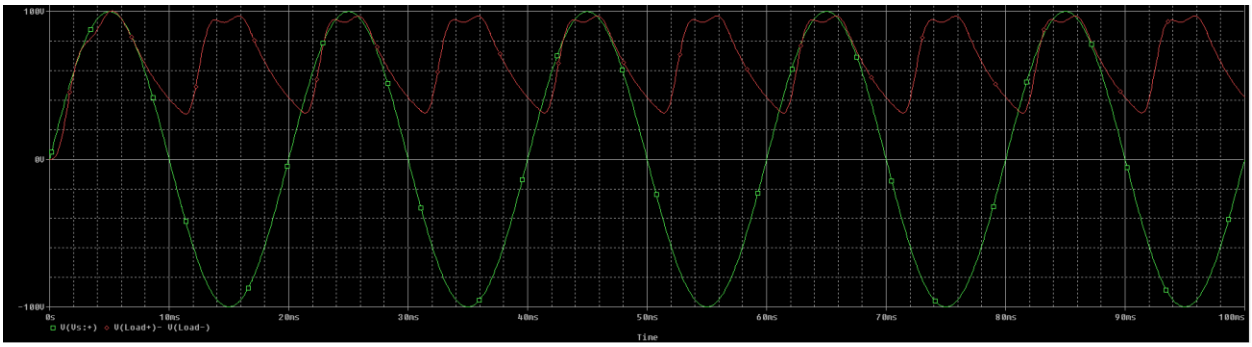
- Plot the output voltages for the three values of the ac side inductance

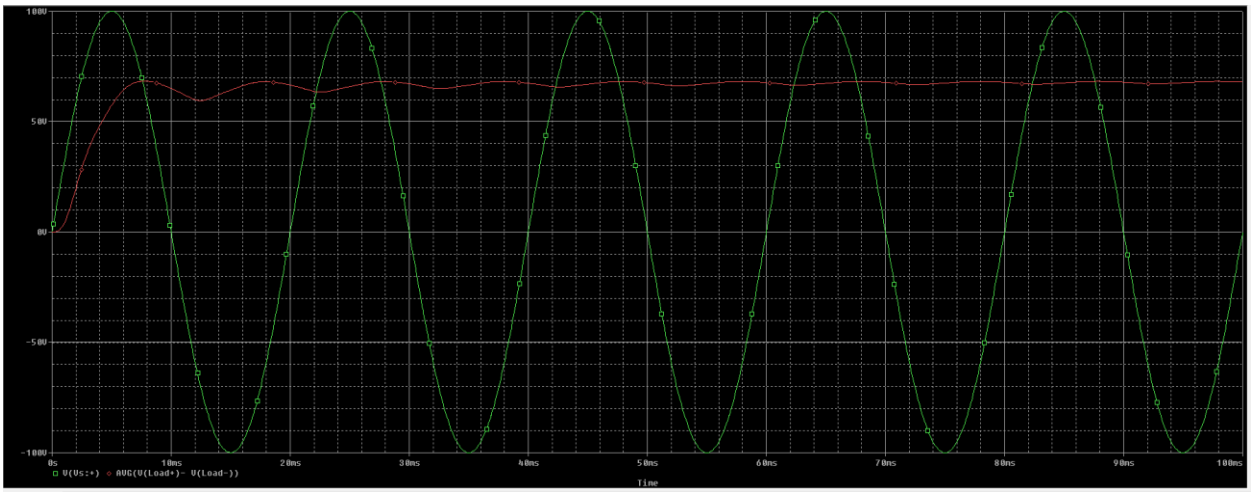
(insert plots here)

- Measure the average values of the output voltages for the three values of ac side inductance

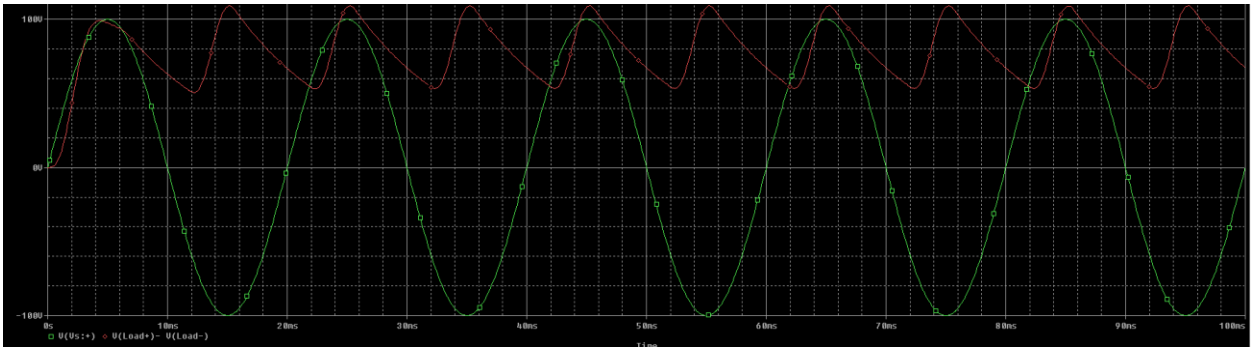
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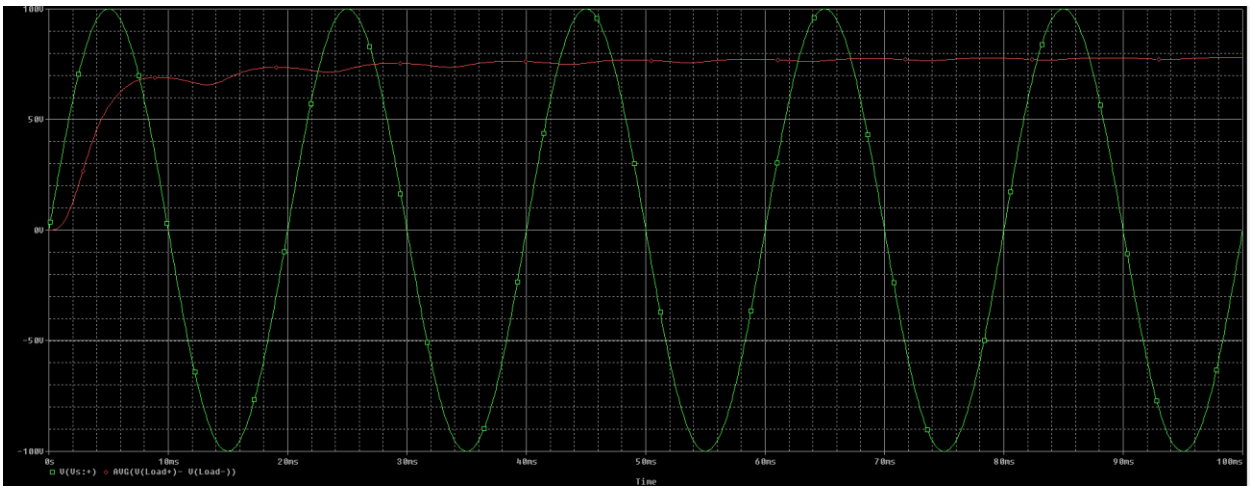
C = 220µF



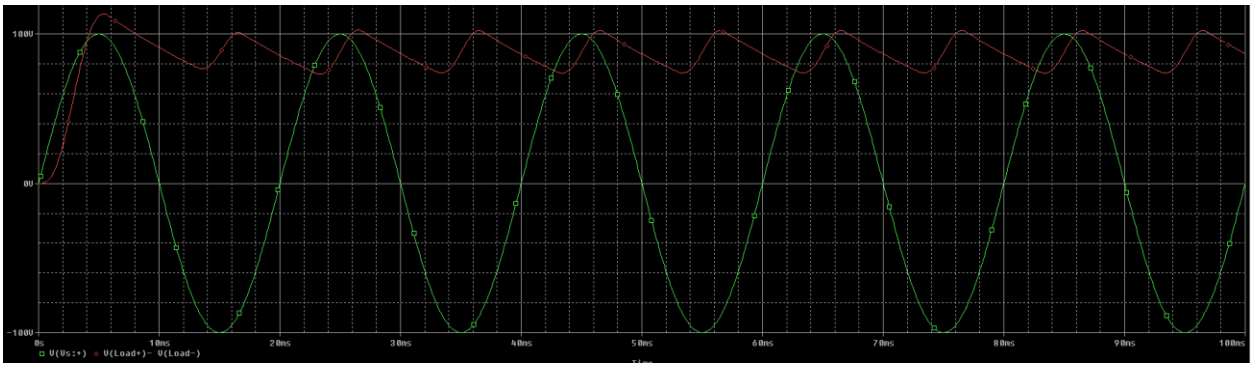


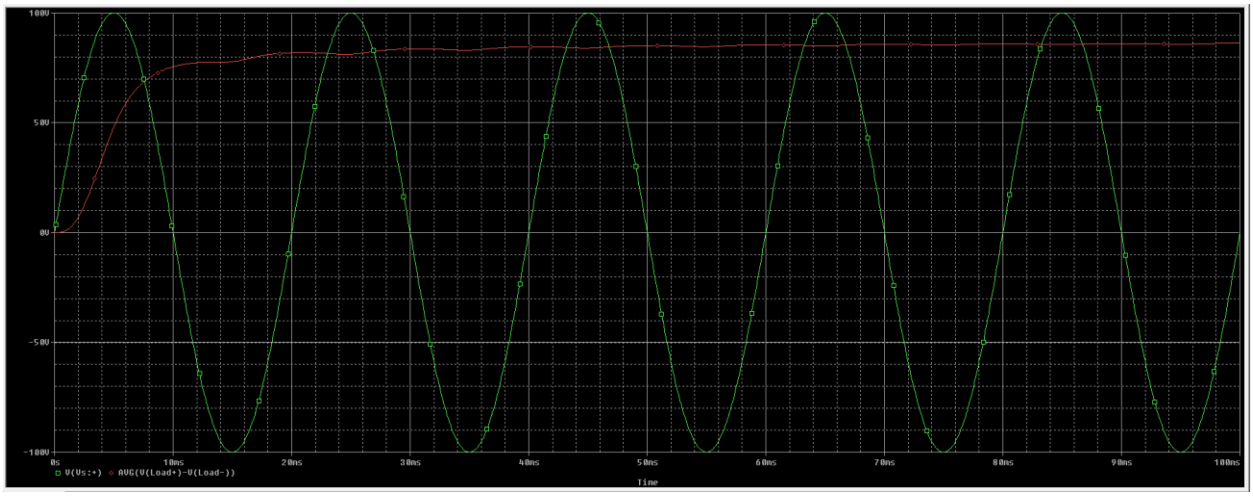
C = 470µF





C = 1000μF





Comment:

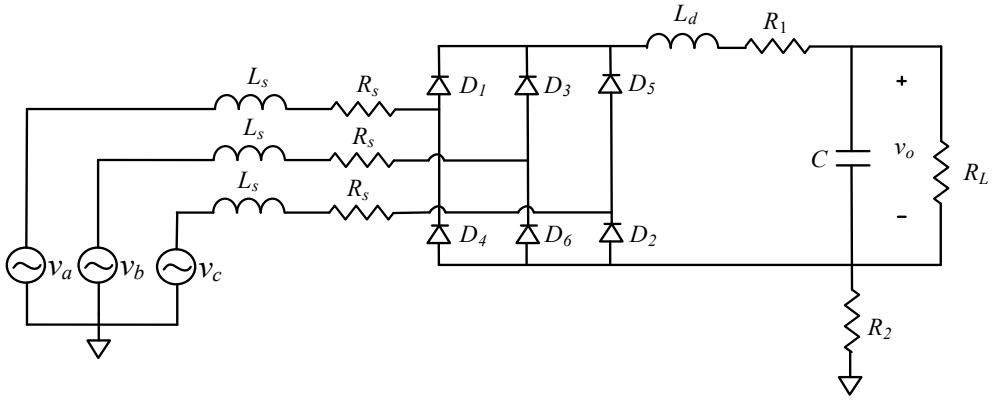
C = 220uF Exhibits the largest ripple, indicating the capacitor discharges significantly during the non-conducting periods of the diode bridge.

C=1000uF Exhibits the smallest ripple. The larger reservoir of charge allows the capacitor to sustain the load voltage more effectively between AC peaks, resulting in a smoother, more constant DC output.

There is a clear inverse relationship between the capacitance value and the peak-to-peak voltage ripple.

Task 2. Three-phase diode bridge rectifier

Objective: The objective of this experiment is to look at the input current and output voltage waveform in the three-phase diode bridge rectifier with large capacitor on the dc side.
Build the circuit as follows:
where
$v_a(t)$ = 100sin(2*pi*50) (V)
$v_b(t)$ = 100sin(2*pi*50-120^0) (V)
$v_c(t)$ = 100sin(2*pi*50+120^0) (V)
L_s = 0.1mH, 0.5mH, 1mH
R_s = 1m
L_d = 0.5mH
R_1 = 5m
C = 500μF
R_L = 20Ω
R_2 = 10000k



a) Plot the input current waveforms for three values of the ac side inductance, L_s = 0.1mH, 0.5mH, 1mH. Comment on the effect of Ls.
(insert plots here)

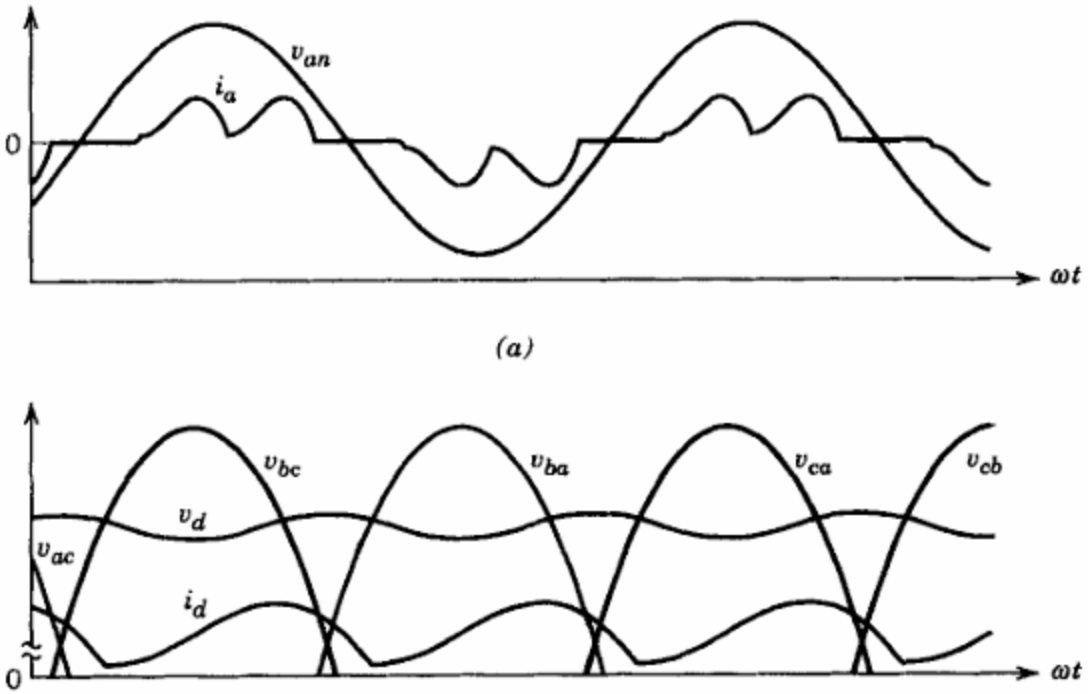
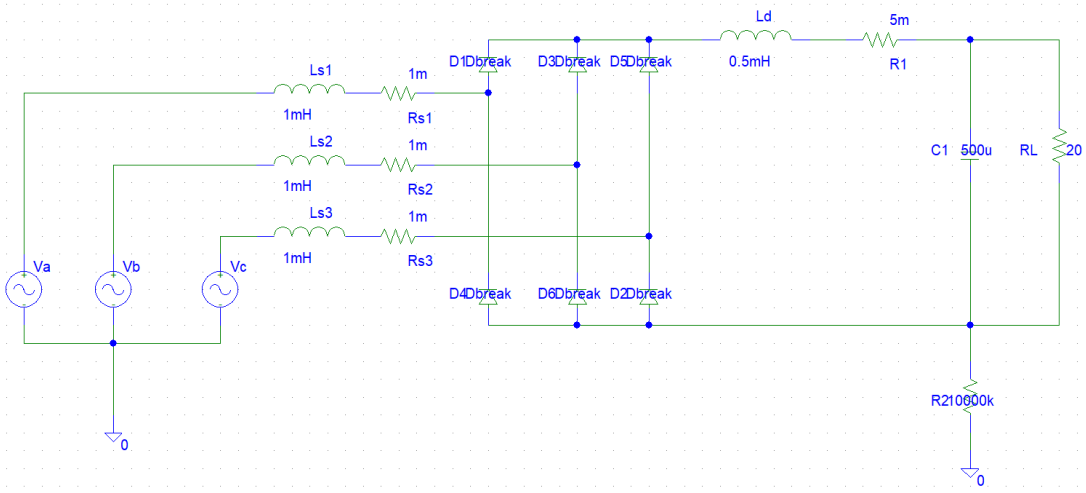
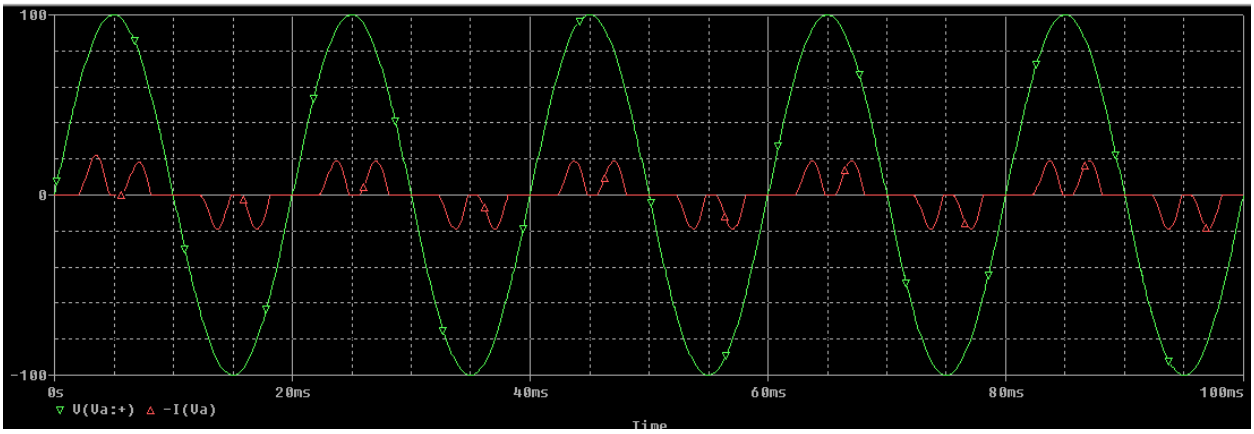


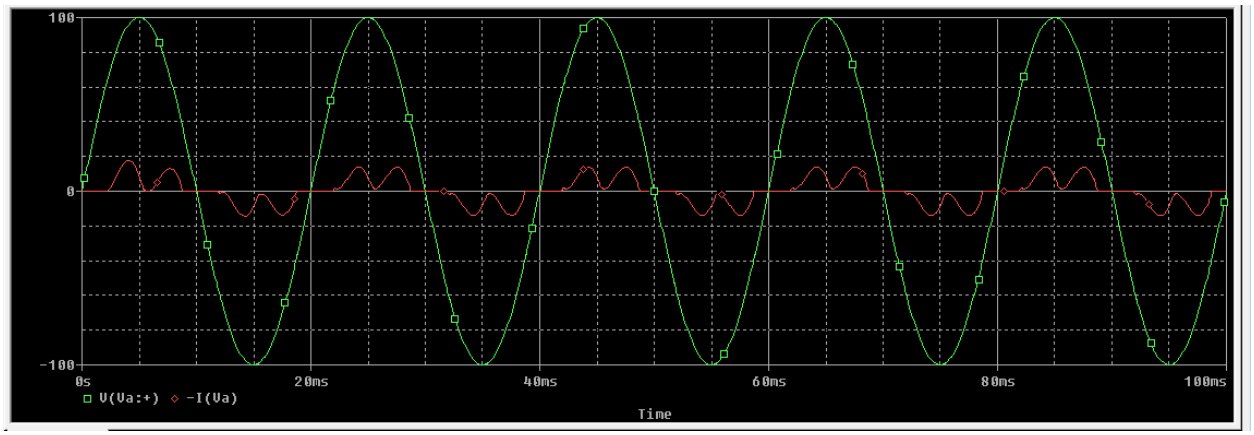
Figure 5-39 Waveforms in the rectifier of Fig. 5-30, obtained in Example 5-7.



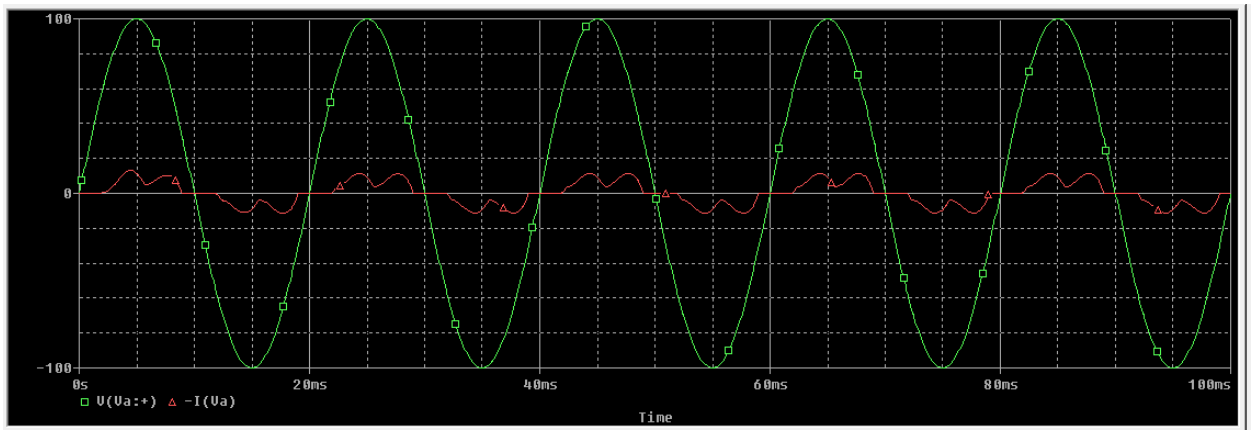
L_s = 0.1mH,



L_s = 0.5mH,



L_s = 1mH



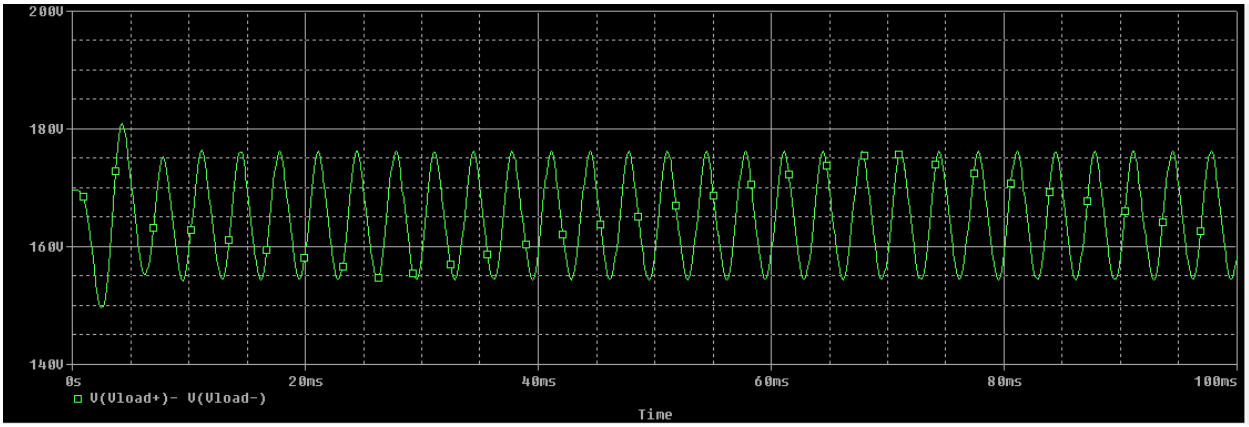
Comment:

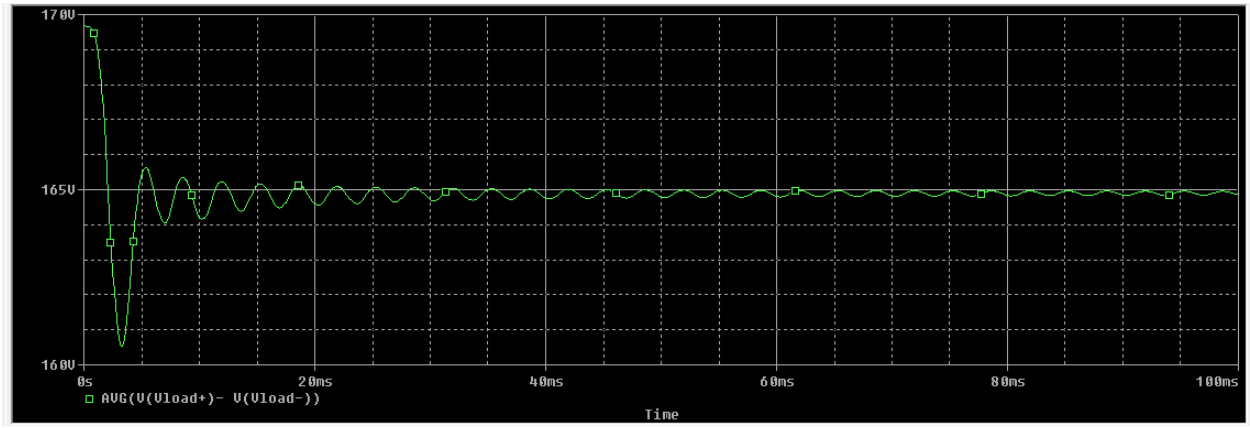
Similar to the single-phase case, increasing Ls acts as a filter. At Ls = 0.1 mH, the current peaks are sharp and high. At L_s = 1mH, the peaks are dampened, the waveform is smoother, and the conduction period for the diodes is extended

b) Plot the output voltage waveforms for three values of the ac side inductance, L_s = 0.1mH, 0.5mH, 1mH.
(insert plots here)

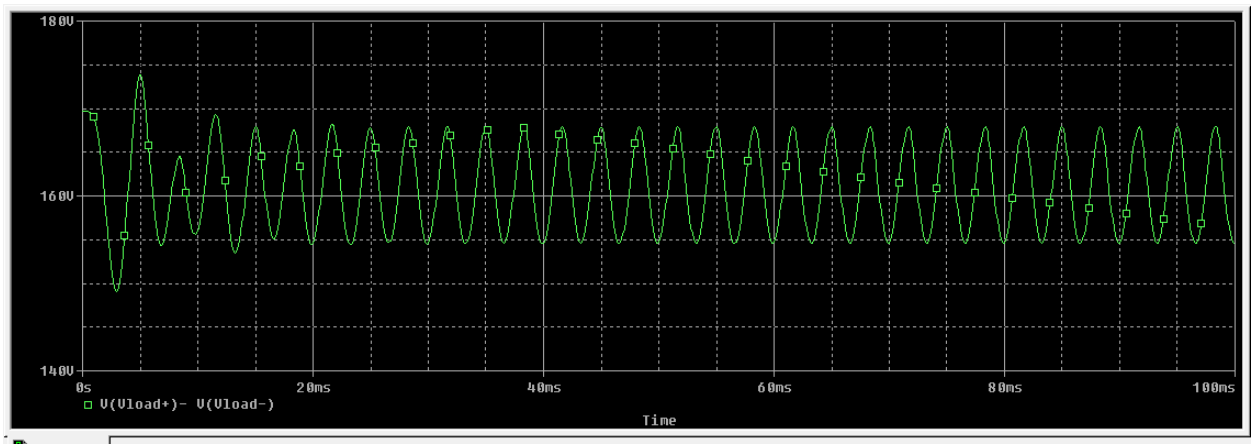
Measure the average values of the output voltages for the three values of ac side inductance (To measure the average values, select “Simulate”, then “Add traces”, and then “AVG()”, type the output variable at the “Trace Expression”)

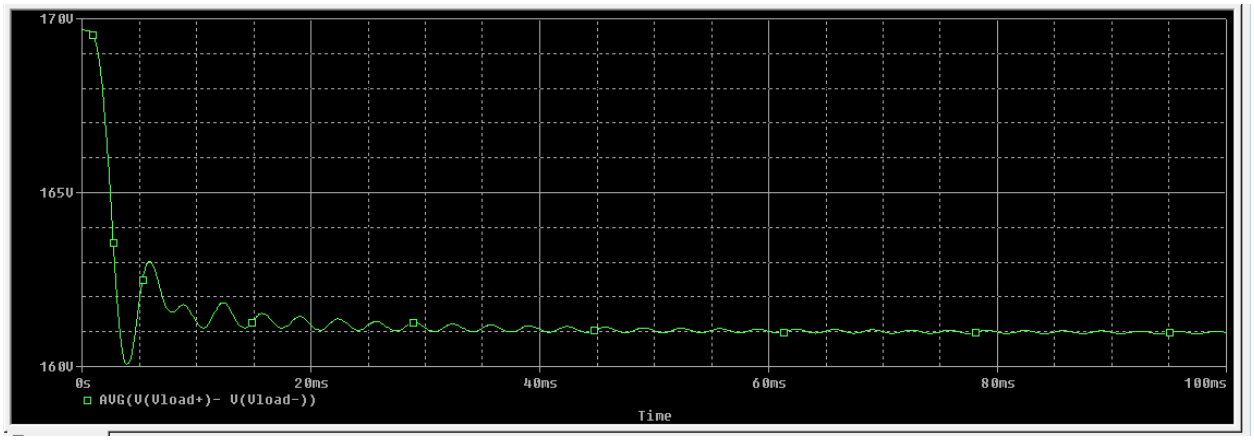
L_s = 0.1mH,



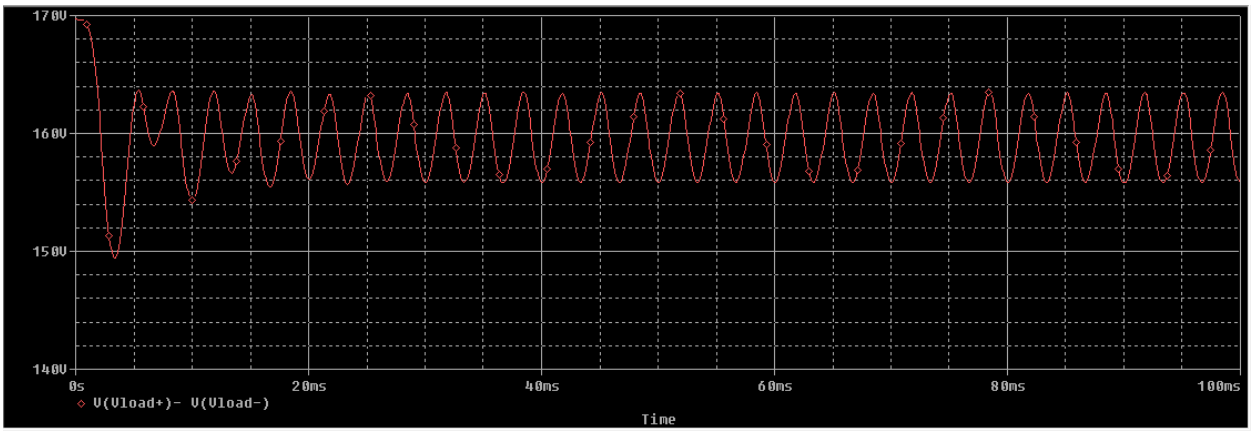


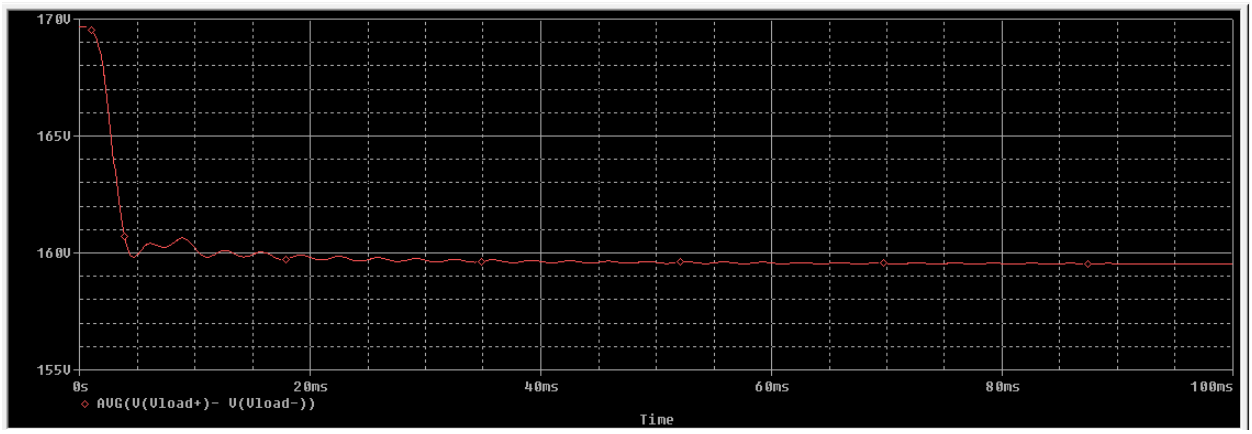
L_s = 0.5mH,


180V
160V
140V
0s
20ms
40ms
60ms
80ms
100ms
V(Vload+)- V(Vload-)
Time


170V
165V
160V
0s
20ms
40ms
60ms
80ms
100ms
AVG(V(Vload+)- V(Vload-))
Time

L_s = 1mH.


170V
160V
150V
140V
0s
20ms
40ms
60ms
80ms
100ms
V(Vload+)- V(Vload-)
Time



Comment:

At **L_s = 0.1 mH**, the average voltage settles near **165V**.

At **L_s = 0.5 mH**, the average voltage settles near **161V**.

At **L_s = 1 mH**, the average voltage drops to roughly **160V**.

The inductance causes a commutation voltage drop..